Education

Cognitive Debt

Cognitive debt is the long term consequence of continuously offloading and delegating thinking to AI. This is clearly demonstrated in MIT's study on the effect of using Large Language Models(LLMs) to produce essays.

The subjects were divided into 3 groups: brain-only, search engine, and LLM usage. They were asked to write essays about a certain topic with their respective tools. At the 4th session, those who were in the brain-only group were moved to the LLM group, and those who were in the LLM group were moved to the brain-only group.

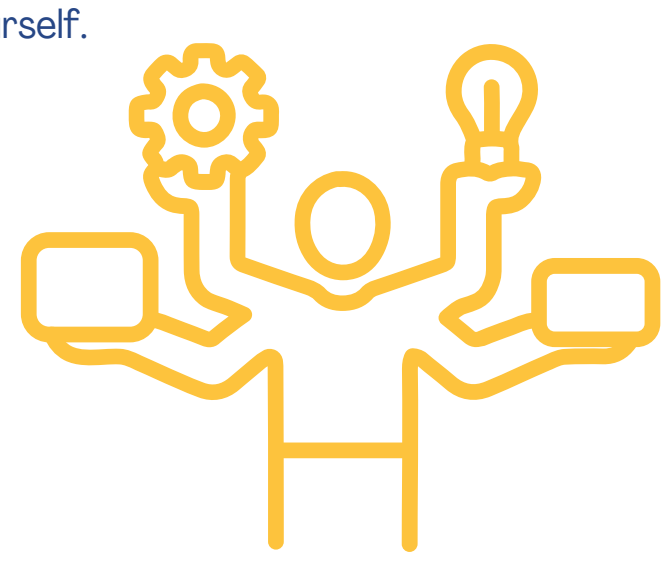
It was found that the group that used LLMs to write their essays had the least memory retention, ownership claiming, and the least connectivity in the brain. However, those in the brain-only group had a consistent, high connectivity between brain tissue, showing high neuron engagement. The search engine group had moderate connections.

The group that went from brain-only to LLMs showed a high level of engagement and memory retention, even when they moved; The group that went from LLMs to brain-only showed the lowest memory retention rates, and struggled to engage meaninfully in the activity.

Over the 4 months that this experiment was conducted, the LLM group showed the lowest performance in neural, linguistic and behavioral areas.

Myths about using AI for academics

Myth The socratic method is how people will use AI to enhance their thinking. We will use AI as active critical thinking partners, enhancing the educational landscape.
Truth Students are always looking for shortcuts. Unless the fundamental behavior of students can be changed, the socratic method will work for the masses only in theory.

Myth AI can be used to save time so that we can use our actual time more valuably.
Truth Using AI requires fact checking because of hallucinations. Going back over all of the information presented by AI takes just as much time as researching and writing yourself.

Myth AI allows us to enhance the skills that we already have.
Truth Artificial Intelligence takes away skills that you already have. The AI may improve your work for now, but in the long term, you will start to lose that skill entirely.